The Universal Explanation Impossibility:
No Explanation Is Faithful, Stable, and Decisive
Under Underspecification

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Abstract

We prove a universal impossibility: no explanation system can simultaneously be faithful (reflecting the chosen hypothesis), stable (consistent across equivalent hypotheses), and decisive (resolving all explanation queries) when the underlying inference problem is underspecified—that is, when multiple hypotheses fit the data equally well. This single abstract theorem, which we call the *Universal Explanation Impossibility*, is a meta-theorem that unifies six apparently unrelated impossibility results from across machine learning: (1) the feature attribution impossibility under collinearity [Bilodeau et al., 2024], (2) the SHAP instability under model multiplicity [D'Amour et al., 2022], (3) the fairness metric incompatibility [Chouldechova, 2017, Kleinberg et al., 2017], (4) the model selection instability under the Rashomon effect [Rudin et al., 2024, Fisher et al., 2019], (5) the causal discovery ambiguity within a Markov equivalence class [Verma and Pearl, 1991], and (6) the impossibility of faithful explanations for overparameterized neural networks. In each case, underspecification induces a Rashomon property on explanation targets, and the same four-line proof closes the impossibility. The resolution is uniform: G-invariant explanation systems that aggregate over equivalent hypotheses restore faithfulness and stability at the cost of decisiveness, and DASH (**D**iversified **A**ggregation of **SH**AP) is a concrete instance of this resolution for feature attribution. The impossibility is ubiquitous: any learning problem with strictly more parameters than constraints admits a Rashomon set of positive measure, and neural networks with hidden dimension exceeding the data dimension always fall into this regime. The framework is mechanically verified in Lean 4 (67 files, 319 theorems, 60 axioms, 0 `sorry`).

1 Introduction

2 The Abstract Framework

We introduce a minimal abstract framework sufficient to state and prove the Universal Explanation Impossibility. All six instances in Section 3 are special cases.

2.1 Explanation Systems

Definition 1 (Explanation System)**.** *An* explanation system *is a tuple* $\mathcal{S} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ *where:*

- $\mathbf{\Theta}$ *is a* parameter space *(the set of possible ground truths or data-generating processes);*

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- $\mathcal{H}$ *is a* hypothesis space *(the set of fitted models or inferred structures); each* $h \in \mathcal{H}$ *is associated with a parameter* $\theta(h) \in \mathbf{\Theta}$ *via the* attribution map $\theta : \mathcal{H} \to \mathbf{\Theta}$*;*
- Y *is an* explanation target space *(e.g., feature rankings, causal graphs, fairness certificates);*
- $\mathsf{obs} : \mathcal{H} \to Y$ *is the* observation map*, assigning to each hypothesis the explanation it "honestly reports";*
- $\mathsf{exp} : \mathcal{H} \to Y$ *is the* explanation map*, the procedure under evaluation; and*
- $\perp \subseteq Y \times Y$ *is an* incompatibility relation *on explanations (e.g., strict ordering* $j \succ k$ *vs.* $k \succ j$*, or two mutually exclusive fairness certificates).*

The explanation map exp is what a practitioner *chooses* (e.g., SHAP, LIME, a fixed causal orientation rule); the observation map obs records what the hypothesis *actually implies*. The incompatibility relation captures when two explanations cannot both be simultaneously correct.

2.2 The Three Desiderata

Definition 2 (Faithful)**.** *An explanation map* exp *is* faithful *if for every hypothesis* $h \in \mathcal{H}$*,*

$$\mathsf{exp}(h) = \mathsf{obs}(h).$$

That is, the produced explanation agrees with what the hypothesis itself implies.

Definition 3 (Stable)**.** *An explanation map* exp *is* stable *if for every pair of hypotheses* $h, h' \in \mathcal{H}$ *with* $\theta(h) = \theta(h')$*,*

$$\mathsf{exp}(h) = \mathsf{exp}(h').$$

That is, equivalent hypotheses (same ground-truth parameter) receive the same explanation.

Definition 4 (Decisive)**.** *An explanation map* exp *is* decisive *if for every hypothesis* $h \in \mathcal{H}$*, the explanation* $\mathsf{exp}(h)$ *is* resolved*: it contains no undecided or tied queries. Formally, there is no* $y \in Y$ *such that both* $(\mathsf{exp}(h), y) \in \perp$ *and* $\mathsf{exp}(h) = y$*.*

In the feature attribution instance (Section 3.1), faithfulness is faithfulness to the model's own SHAP values, stability is consistency across retraining seeds, and decisiveness is the requirement to rank every feature pair strictly (no ties).

2.3 The Rashomon Property

The key structural condition linking underspecification to explanation impossibility is the following.

Definition 5 (Rashomon Property)**.** *An explanation system* $\mathcal{S}$ *has the* Rashomon property *if for every resolved pair of incompatible explanations* $(y, y') \in \perp$ *with* $y \neq y'$*, there exist hypotheses* $h, h' \in \mathcal{H}$ *with* $\theta(h) = \theta(h')$ *such that*

$$\mathsf{obs}(h) = y \quad \textit{and} \quad \mathsf{obs}(h') = y'.$$

That is, for every incompatible explanation pair, two hypotheses with the same ground-truth parameter exist that honestly *report the respective incompatible explanations.*

The Rashomon property is the formal statement of underspecification at the explanation level: the data do not uniquely determine the explanation, even in principle. It generalizes the Rashomon effect of Rudin et al. [2024], Fisher et al. [2019] from model selection to arbitrary explanation targets.

2.4 Main Theorem

Theorem 6 (Universal Explanation Impossibility)**.** *If an explanation system* $\mathcal{S} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ *has the Rashomon property, then no explanation map* exp *can simultaneously be faithful, stable, and decisive.*

Proof. By the Rashomon property, there exist incompatible explanations $y \neq y'$ with $(y, y') \in \perp$ and hypotheses h, h' satisfying $\theta(h) = \theta(h')$, $\mathsf{obs}(h) = y$, and $\mathsf{obs}(h') = y'$.

Suppose for contradiction that exp is faithful, stable, and decisive. By faithfulness, $\mathsf{exp}(h) = \mathsf{obs}(h) = y$ and $\mathsf{exp}(h') = \mathsf{obs}(h') = y'$. But by stability and $\theta(h) = \theta(h')$, we must have $\mathsf{exp}(h) = \mathsf{exp}(h')$, i.e., $y = y'$. This contradicts $y \neq y'$. □

Remark 7. *The proof uses decisiveness only implicitly, through the requirement that* $\mathsf{obs}(h)$ *is a resolved, specific explanation (not a set or distribution). The impossibility persists whenever* obs *is single-valued and the incompatibility relation is asymmetric.*

Remark 8. *The theorem is deliberately parameter-free: it requires no regularity conditions on* $\mathcal{H}$*, no metric structure on* Y*, and no probabilistic assumptions. All quantitative depth (divergence rates, sample complexity, ensemble sizes) appears in the instance-specific sections below.*

Corollary 9 (Decidable Underspecification Implies Impossible Explanation)**.** *If the data-generating process* θ *is not identifiable from the observations—i.e., if there exist* h, h' *with* $\theta(h) = \theta(h')$ *yet* $\mathsf{obs}(h) \neq \mathsf{obs}(h')$*—then the Rashomon property holds whenever the explanation targets are ordered by an incompatibility relation with at least one incompatible pair.*

3 Six Instances

We instantiate Theorem 6 in six structurally distinct domains. In each case we (i) specify the explanation system $(\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$, (ii) verify the Rashomon property, and (iii) characterize the resolution.

3.1 Instance 1: Feature Attribution Under Collinearity

We instantiate the abstract framework (Definition 1) for feature attribution, recovering the Attribution Impossibility of the companion paper as a corollary of Theorem 6.

The explanation system. Fix $P \geq 2$ features partitioned into L collinearity groups. Within each group $\ell \in [L]$, every pair of features shares pairwise correlation $\rho \in (0, 1)$; features across groups are independent. The explanation system $\mathcal{S}_{\mathrm{attr}} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ is:

- $\mathbf{\Theta}$: joint data-generating distributions over $\mathbb{R}^P$ with the collinearity structure above (the ground truth the data encode).
- $\mathcal{H}$: trained models f from a fixed model class (e.g. GBDT, Lasso, neural networks), indexed by random training seed; each f is associated with its data-generating distribution $\theta(f) \in \mathbf{\Theta}$ via the attribution map $\theta : \mathcal{H} \to \mathbf{\Theta}$.
- $Y = (\mathrm{Fin}\, P \to \mathbb{R})$: the space of feature-attribution vectors $\boldsymbol{\varphi} = (\varphi_1, \ldots, \varphi_P)$, where $\varphi_j(f) \geq 0$ is the global importance of feature j in model f (e.g. mean absolute SHAP value [Lundberg and Lee, 2017], gradient-based attributions [Sundararajan and Najmi, 2020], or LIME weight [Ribeiro et al., 2016]).
- $\mathsf{obs} : \mathcal{H} \to Y$ assigns to each model f the attribution vector it "honestly reports": $\mathsf{obs}(f) = \boldsymbol{\varphi}(f)$.
- $\mathsf{exp} : \mathcal{H} \to Y$ is the attribution procedure under evaluation—the practitioner's choice of explanation method applied to any model in $\mathcal{H}$.
- $\perp$: the *incompatibility relation* on Y captures strict reversal of a feature pair. Formally, $\boldsymbol{\varphi} \perp \boldsymbol{\varphi}'$ if there exists a group ℓ and distinct features $j, k \in \ell$ such that

$$\varphi_j > \varphi_k \quad \text{and} \quad \varphi'_k > \varphi'_j.$$

  Two attribution vectors are incompatible when they rank the same collinear pair in opposite orders; both orderings cannot simultaneously be correct.

Rashomon property. Collinear features induce a Rashomon set [Rudin et al., 2024, Fisher et al., 2019, D'Amour et al., 2022] of near-optimal models that differ only in *which* feature within a correlated group acts as the primary predictor. Formally, $\mathcal{S}_{\mathrm{attr}}$ has the Rashomon property (Definition 5): for any group ℓ and any distinct features $j, k \in \ell$, there exist models $f, f' \in \mathcal{H}$ with $\theta(f) = \theta(f')$ (same data-generating distribution) such that

$$\varphi_j(f) > \varphi_k(f) \qquad \text{and} \qquad \varphi_k(f') > \varphi_j(f').$$

This is the `RashimonProperty` of `Trilemma.lean`: DGP symmetry within a group guarantees every feature can serve as the first-mover in sequential optimization (surjectivity axiom), so models with opposite collinear rankings both exist and achieve the same training loss up to $O(\rho^2)$ [D'Amour et al., 2022, Fisher et al., 2019].

Definition 10 (Incompatible attribution rankings)**.** *Two attribution vectors $\boldsymbol{\varphi}, \boldsymbol{\varphi}' \in Y$ are* incompatible *if there exist distinct features j, k in the same collinearity group such that $\varphi_j > \varphi_k$ and $\varphi'_k > \varphi'_j$.*

Instance 1 of the Universal Theorem.

Corollary 11 (Attribution Impossibility)**.** *The explanation system $\mathcal{S}_{\mathrm{attr}}$ has the Rashomon property. Therefore, by Theorem 6, no attribution map* exp *can simultaneously be faithful (attributions agree with the model's own values), stable (same attributions regardless of which equivalent model is explained), and decisive (every collinear pair is strictly ranked).*

Proof. The Rashomon property was established above. The impossibility is then a direct application of Theorem 6 to $\mathcal{S}_{\mathrm{attr}}$. □

In the notation of the companion paper (the Attribution Impossibility paper), faithfulness corresponds to faithfulness (Definition 2), stability to stability (Definition 3), and decisiveness to completeness (every feature pair is strictly ranked). Corollary 11 is the abstract-framework restatement of the companion Attribution Impossibility (Theorem 1 of the companion paper), verified as `attribution_impossibility` in `Trilemma.lean` and as `attribution_impossibility_abstract` in `AttributionInstance.lean` (zero model-specific axiom dependencies in both cases).

Quantitative depth. The abstract impossibility is qualitative. Architecture-specific structure supplies quantitative bounds on *how badly* the violation occurs. For GBDT with per-group correlation ρ, the attribution ratio (dominant versus symmetric partner) diverges as $1/(1-\rho^2)$ as $\rho \to 1$ (Lean-verified in `Ratio.lean`; see also D'Amour et al. [2022]). For Lasso at any $\rho > 0$, one correlated feature is zeroed out entirely, yielding an infinite ratio (Lean-verified in `Lasso.lean`). For random forests with T trees, ensemble averaging drives the ratio to $O(1/\sqrt{T})$—the lone converging case and the mechanism behind DASH.

Resolution. The G-invariant resolution (Section 4) for this instance is DASH (**D**iversified **A**ggregation of **SH**AP): averaging attribution vectors over an ensemble of retrained models collapses the Rashomon set to a single equitable vector with $\mathbb{E}[\varphi_j] = \mathbb{E}[\varphi_k]$ for symmetric features j, k, restoring faithfulness (in expectation) and stability at the cost of decisiveness—collinear pairs are tied rather than arbitrarily ranked.

3.2 Instance 2: Attention Maps as Explanations

Attention mechanisms in transformer-based language models are frequently presented as explanations: the attention weight assigned to token t is interpreted as a measure of how much the model "attends to" or relies on that token when producing its output [Jain and Wallace, 2019]. We show that this interpretation is incompatible with the three desiderata under underspecification.

The explanation system. Let T denote the number of tokens in the context. We instantiate the general framework as the 6-tuple $\mathcal{S}_{\mathrm{attn}} = (\boldsymbol{\Theta}_{\mathrm{attn}}, \mathcal{H}_{\mathrm{attn}}, Y_{\mathrm{attn}}, \mathsf{obs}_{\mathrm{attn}}, \mathsf{exp}_{\mathrm{attn}}, \perp_{\mathrm{attn}})$ where:

- $\boldsymbol{\Theta}_{\mathrm{attn}}$ is the space of neural network weight configurations θ (query, key, value, and output projection matrices at every layer and head);
- $\mathcal{H}_{\mathrm{attn}} = \Delta^T$ is the probability simplex over T tokens—the space of attention distributions $\alpha \in \mathbb{R}^T$ with $\alpha_t \geq 0$ and $\sum_{t=1}^{T} \alpha_t = 1$;
- Y_{attn} is the space of input-to-output functions $f : \mathcal{X} \to \mathcal{Y}$, i.e., the observable input→output behavior of the network;
- $\mathsf{obs}_{\mathrm{attn}}(\theta) = f_\theta$ is the function computed by the network with weights θ;

- $\mathsf{exp}_{\mathrm{attn}}(\theta) = \alpha_\theta \in \Delta^T$ is the attention distribution produced by θ on a fixed input $x \in \mathcal{X}$; and
- $(\alpha, \alpha') \in \perp_{\mathrm{attn}}$ if and only if $\mathrm{argmax}_t\, \alpha_t \neq \mathrm{argmax}_t\, \alpha'_t$, i.e., the two distributions assign peak attention to different tokens.

The Rashomon property. D'Amour et al. [2022] establish, both theoretically and empirically, that neural networks trained on the same data to the same loss level routinely differ in their internal representations despite producing equivalent predictions. Applied to attention, this yields the Rashomon property for $\mathcal{S}_{\mathrm{attn}}$: there exist weight configurations θ_1, θ_2 such that

$$f_{\theta_1} \approx f_{\theta_2} \quad \text{(equivalent predictions, same test loss)} \quad \text{yet} \quad \mathrm{argmax}_t(\alpha_{\theta_1})_t \neq \mathrm{argmax}_t(\alpha_{\theta_2})_t.$$

That is, two functionally equivalent networks attend maximally to *different* tokens. This is not an artifact of optimization noise: Jain and Wallace [2019] show directly that attention weights can be dramatically permuted—including adversarially—while leaving model predictions unchanged, providing a sharp empirical demonstration that attention is not a faithful explanation in the sense of Definition 2.

Instance 2: Attention Impossibility.

Theorem 12 (Attention Map Impossibility)**.** *The attention explanation system $\mathcal{S}_{\mathrm{attn}}$ has the Rashomon property. Therefore, by Theorem 6, no attention map $\mathsf{exp}_{\mathrm{attn}}$ can simultaneously be faithful (report the network's actual attention), stable (assign the same dominant token to equivalent networks), and decisive (commit to a unique most-attended token).*

Proof. By the Rashomon property, there exist θ_1, θ_2 with $\mathsf{obs}_{\mathrm{attn}}(\theta_1) = \mathsf{obs}_{\mathrm{attn}}(\theta_2)$ (same input→output function) and $(\alpha_{\theta_1}, \alpha_{\theta_2}) \in \perp_{\mathrm{attn}}$ (different argmax tokens). The result then follows immediately from Theorem 6. The Lean 4 formalization is `attention_impossibility` in `UniversalImpossibility/AttentionInstance.lean`. □

Implications for the attention-as-explanation debate. Theorem 12 gives a structural resolution to the debate initiated by Jain and Wallace [2019] and Wiegreffe and Pinter [2019]. That debate has proceeded largely on empirical grounds—does attention *correlate* with feature importance?—but the impossibility is not contingent on correlation structure. Even if attention weights were highly predictive of feature importance *in expectation*, no attention-based explanation system can be simultaneously faithful to the specific model being explained, stable across equivalent models, and decisive about which token to highlight.

The trilemma has a concrete consequence for practitioners: any attention visualization that commits to a single most-attended token (decisive) must either be unfaithful to some equivalent models or change its output when the model is retrained from a different random seed. This is not a failure of implementation but a mathematical inevitability whenever two equally predictive networks disagree on their peak-attention token.

The resolution prescribed by Section 4 applies directly: a G-invariant attention explanation aggregates attention distributions across the Rashomon set of equivalent models, yielding an averaged distribution $\bar{\alpha} = \frac{1}{|\mathcal{R}|} \sum_{\theta \in \mathcal{R}} \alpha_\theta$ that is faithful in expectation and stable by construction, but typically fails to be decisive (the averaged distribution may spread mass across multiple tokens, refusing to crown a single "most-attended" token). Reporting $\bar{\alpha}$ rather than α_θ is the attention analog of reporting CPDAG edge confidences rather than a single causal orientation—an honest acknowledgment that the data do not uniquely determine an attention explanation.

3.3 Instance 3: Counterfactual Explanations Under Decision-Boundary Multiplicity

Counterfactual explanations answer the question: "What is the smallest change to my input that would flip the model's prediction?" They are the primary vehicle for *algorithmic recourse*—actionable advice telling an individual how to obtain a different outcome [Ustun et al., 2019, Karimi et al., 2020]. We show that no counterfactual explanation method can simultaneously be faithful, stable, and decisive when multiple models fit the data equally well.

The explanation system. We instantiate the abstract framework (Definition 1) as follows.

- $\mathbf{\Theta} = \mathbf{\Theta}$: the space of *model parameters*, each $\theta \in \mathbf{\Theta}$ defining a decision boundary $\partial f_\theta = \{x \in \mathcal{X} : f_\theta(x) = \tau\}$ in the input space $\mathcal{X} \subseteq \mathbb{R}^P$ for a threshold τ.
- $\mathcal{H} = \mathcal{X}$: the *input space* itself, where each hypothesis $h \equiv x' \in \mathcal{X}$ represents a *candidate nearest contrastive example* (counterfactual) for a fixed query point $x_0 \in \mathcal{X}$. The attribution map $\theta : \mathcal{H} \to \mathbf{\Theta}$ sends x' to the model parameters for which x' is the nearest point with a different prediction.
- Y: the space of *prediction vectors* on a held-out test set, i.e., $Y = \{0, 1\}^n$ for n test points. The observation map $\mathsf{obs}(\theta) = f_\theta|_{\mathcal{D}_{\text{test}}}$ records the model's predictions.

The *explanation map* $\mathsf{exp}(\theta) = \arg\min_{x' : f_\theta(x') \neq f_\theta(x_0)} \|x' - x_0\|$ is the nearest counterfactual for query x_0 under model θ. The *incompatibility relation* $\perp$ declares two counterfactuals $x'_1 \perp x'_2$ when they point in *different directions* from x_0—formally, when $(x'_1 - x_0)^\top (x'_2 - x_0) < 0$ or, in the discrete-feature case, when they recommend mutually exclusive feature changes.

The Rashomon property. The Rashomon effect for decision boundaries is well-documented: within a small loss tolerance ϵ, an exponential number of models may agree on test predictions while placing their decision boundaries in qualitatively different locations [Semenova et al., 2022, Fisher et al., 2019]. Formally, the Rashomon set $\mathcal{R}_\epsilon = \{\theta \in \mathbf{\Theta} : \mathcal{L}(\theta) \leq \mathcal{L}(\theta^*) + \epsilon\}$ generically contains models whose boundaries straddle the query point x_0 from opposite sides. Concretely:

- **Model 1** (θ_1): decision boundary passes *above* x_0 in the relevant direction; the nearest counterfactual is x'_1 (move feature j up).
- **Model 2** (θ_2): decision boundary passes *below* x_0; the nearest counterfactual is x'_2 (move feature j down).

Both θ_1 and θ_2 achieve identical test-set predictions ($\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$), yet $x'_1 \perp x'_2$. This is precisely the Rashomon property (Definition 5).

Proposition 13 (Instance 3: Counterfactual Impossibility)**.** *Let $\mathcal{S}_{\text{CF}}$ be the counterfactual explanation system above. If $\mathcal{S}_{\text{CF}}$ has the Rashomon property—i.e., the Rashomon set $\mathcal{R}_\epsilon$ contains models with incompatible nearest counterfactuals for some query x_0—then no counterfactual explanation map* exp *can simultaneously be faithful, stable, and decisive.*

Proof. Direct application of Theorem 6 to $\mathcal{S}_{\text{CF}}$. The Rashomon property supplies hypotheses θ_1, θ_2 with $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ and $\mathsf{obs}_{\text{CF}}(\theta_1) \perp \mathsf{obs}_{\text{CF}}(\theta_2)$. The four-line proof of Theorem 6 then yields a contradiction from faithfulness, stability, and decisiveness simultaneously. This is mechanically verified as `counterfactual_impossibility` in `UniversalImpossibility/CounterfactualInstance.lean`. □

Implications for recourse and actionable explanations. Proposition 13 reveals a fundamental limit on the reliability of algorithmic recourse. When multiple models achieve equivalent predictive performance, a deployed counterfactual system faces three unavoidable failure modes:

1. **Recourse instability.** An individual who retrains a lender's model (e.g., after a software update) may receive incompatible advice—"increase savings" vs. "reduce debt"—despite the model's credit-scoring accuracy being unchanged. Empirical evidence of this instability is documented by Marx et al. [2024].
2. **Fragility under model multiplicity.** Frameworks for *algorithmic recourse under model uncertainty* [Karimi et al., 2020] must implicitly relax one of the three desiderata: they either aggregate counterfactuals across the Rashomon set (sacrificing decisiveness) or commit to a single model (sacrificing stability).
3. **Legal compliance risk.** Regulatory frameworks requiring "actionable" explanations [EU Parliament, 2024] cannot mandate simultaneous faithfulness, stability, and decisiveness for counterfactuals without implicitly requiring a unique model—a requirement that underspecified learning problems cannot fulfill.

The uniform resolution (Section 4) applies here: a G-invariant counterfactual system averages the nearest counterfactual direction over the Rashomon set, producing a *robust recourse direction* that is faithful in expectation and stable, at the cost of yielding a distribution over directions rather than a single decisive recommendation. Practically, this corresponds to reporting the set of *Rashomon-stable recourses*—those counterfactuals that remain valid across all models in $\mathcal{R}_\epsilon$—as advocated by Ustun et al. [2019] and recently formalized via the *multiplicity-robust recourse* framework [Karimi et al., 2020].

3.4 Instance 4: Concept Probe Impossibility

Motivation. Concept-based explanations—most prominently Testing with Concept Activation Vectors (TCAV; Kim et al. 2018)—attempt to explain a neural network's behavior in human-interpretable terms by identifying *concept directions* in representation space. For example, the direction corresponding to "stripes" in an image classifier's penultimate layer is found by fitting a linear probe that separates activations on striped versus non-striped inputs. This direction is then interpreted as the network's internal encoding of the concept, and the degree to which prediction scores change along it is reported as the explanation.

The hope is that concept directions are a stable, faithful window into model internals. We show this hope is mathematically impossible under underspecification: two models with identical predictions can encode the same concept in directions that are genuinely incompatible, so no concept probe can simultaneously be faithful, stable, and decisive.

Instantiation. We instantiate Definition 1 as follows.

Definition 14 (Concept Probe Explanation System)**.** *Let $c \in \mathbb{N}$ be the dimension of the representation space. The* concept probe explanation system *is the tuple $\mathcal{S}_{\mathrm{cp}} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ where:*

- $\mathbf{\Theta} = \Theta$ *is the space of model weight configurations, each $\theta \in \Theta$ specifying a neural network together with its internal representations;*
- $\mathcal{H} = \mathbb{R}^c$ *is the space of* concept activation vectors*—unit directions in the c-dimensional representation space that a linear probe associates with a target concept;*
- Y *is the space of predictions (input-output functions $f_\theta : \mathcal{X} \to \mathcal{Y}$), serving as the observable behavior of the model;*
- $\mathsf{obs}(\theta) = f_\theta$ *is the* observation map*, recording the input-output function computed by weights θ;*
- $\mathsf{exp}(\theta) = v_\theta \in \mathbb{R}^c$ *is the* explanation map*, returning the concept activation direction extracted by a linear probe on the representations of θ; and*
- $(v, v') \in \perp$ *iff v and v' are not related by a rotation within the concept subspace—that is, they designate structurally distinct directions that cannot both be reported as* the *concept direction without contradiction.*

The Rashomon Property for concept probes. The empirical literature provides direct evidence that the Rashomon property holds for concept directions. Bolukbasi et al. [2016] demonstrate that gender directions vary substantially across word-embedding architectures trained on the same corpus: models with identical downstream task performance encode gender in directions that differ in orientation, dimension, and linearity. Kim et al. [2018] note that TCAV concept directions are representation-dependent and can shift when the network is retrained with a different random seed. Ravfogel et al. [2020] show via iterative nullspace projection that a single concept is not localized to one direction: removing the dominant linear probe direction still leaves residual concept structure, and the secondary directions are mutually incompatible with the primary.

Formally, we require:

Definition 15 (Rashomon Property for Concept Probes)**.** *There exist weight configurations $\theta_1, \theta_2 \in \Theta$ such that*

$$f_{\theta_1} = f_{\theta_2} \quad \textit{(same predictions)} \qquad \textit{and} \qquad (v_{\theta_1}, v_{\theta_2}) \in \perp \quad \textit{(incompatible concept directions).}$$

This is the formal content of Definition 5 in $\mathcal{S}_{\mathrm{cp}}$: equivalent models (same observable behavior, hence $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$) report incompatible concept directions. The condition holds in any over-parameterized network class—the zero-loss manifold has dimension $\geq d - n > 0$ for hidden dimension $d > n$ training points, so there are infinitely many weight configurations with identical input-output behavior but geometrically distinct internal representations.

The concept probe impossibility.

Theorem 16 (Concept Probe Impossibility)**.** *If the concept probe explanation system $\mathcal{S}_{\mathrm{cp}}$ has the Rashomon property, then no explanation map* exp *can simultaneously be faithful, stable, and decisive.*

Proof. Immediate from Theorem 6 applied to $\mathcal{S}_{\mathrm{cp}}$. Concretely: let θ_1, θ_2 witness the Rashomon property, so $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ and $(v_{\theta_1}, v_{\theta_2}) \in \perp$. Faithfulness forces $\mathsf{exp}(\theta_i) = v_{\theta_i}$; decisiveness requires each $\mathsf{exp}(\theta_i)$ to be a specific direction; stability forces $\mathsf{exp}(\theta_1) = \mathsf{exp}(\theta_2)$. But then $v_{\theta_1} = v_{\theta_2}$, contradicting $(v_{\theta_1}, v_{\theta_2}) \in \perp$. □

The Lean 4 formalization of this instance is in `UniversalImpossibility/ConceptInstance.lean`. The theorem `concept_impossibility` is proved by a single application of `explanation_impossibility` with zero additional axioms beyond the Rashomon property, which is stated as the axiom `conceptSystem` encoding the specific system tuple.

Implications for probing-as-understanding. Theorem 16 has direct consequences for the practice of using concept probes as explanations of model internals.

1. *Concept directions are not model-invariant facts.* A TCAV direction is a fact about a specific weight configuration, not about the function the network computes. Two retraining runs that produce functionally identical models (same accuracy, same predictions on held-out data) may produce concept directions that are not related by rotation and therefore cannot be reconciled into a single explanation.
2. *Faithfulness and stability cannot coexist.* Requiring a probe to faithfully report the concept direction of the model it was fitted to forces instability across retraining seeds. Requiring stability forces the probe to report a direction that is unfaithful to at least one model in the Rashomon set.
3. *The resolution entails indecisiveness.* The G-invariant resolution (Section 4) applied here would average concept directions across equivalent models, yielding a distribution over directions or a subspace rather than a single vector. This restores faithfulness-in-expectation and stability at the cost of decisiveness: the explanation no longer commits to a single direction, which undermines the interpretive clarity that motivates concept probing in the first place.

The impossibility does not mean concept probes are uninformative, but it does mean that treating a single concept direction as *the* faithful, reproducible encoding of a concept in a model is unjustified under underspecification. Practitioners should report concept directions together with their variability across the Rashomon set—analogously to how DASH reports SHAP values with their ensemble distribution rather than as a single point estimate.

3.5 Instance 5: Causal Discovery Within a Markov Equivalence Class

We instantiate Theorem 6 for causal structure learning, where the fundamental obstruction is Markov equivalence: distinct directed acyclic graphs (DAGs) can be observationally indistinguishable yet disagree on causal edge directions.

The explanation system. Define the 6-tuple $\mathcal{S}_{\mathrm{causal}} = (\mathbf{\Theta},\ \mathcal{H},\ Y,\ \mathsf{obs},\ \mathsf{exp},\ \perp)$ as follows.

- $\mathbf{\Theta}$ is the set of DAGs over a fixed vertex set V (the space of possible causal structures, i.e., data-generating processes).

- $\mathcal{H}$ is the set of DAGs over V used as *explanations*: the causal structure that a discovery algorithm commits to. The attribution map $\theta : \mathcal{H} \to \mathbf{\Theta}$ sends each explanation DAG to the underlying data-generating DAG it was fitted to.
- Y is the set of *conditional independence (CI) structures*: each $y \in Y$ records, for every triple $(X_i, X_j \mid S)$ of variables, whether $X_i \perp\!\!\!\perp X_j \mid S$ holds in the associated distribution. Equivalently, Y indexes Markov equivalence classes.
- $\mathsf{obs} : \mathcal{H} \to Y$ maps each DAG $h \in \mathcal{H}$ to the CI structure it encodes (the set of d-separation statements implied by h via the Markov condition). Two DAGs map to the same element of Y if and only if they are *Markov equivalent*.
- $\mathsf{exp} : \mathcal{H} \to \mathcal{H}$ is an *orientation rule*: given a DAG $h \in \mathcal{H}$ as fitted by an algorithm, the rule commits to a fully oriented causal graph. In practice, exp is any causal discovery procedure (PC algorithm, GES, FCI, etc.) that returns a single DAG rather than an equivalence class.
- $\perp \subseteq \mathcal{H} \times \mathcal{H}$ is the *edge-reversal incompatibility*: $(h, h') \in \perp$ if there exists a directed edge $X \to Y$ in h and $Y \to X$ in h' (or vice versa). Two causal explanations are incompatible when they assert opposite orientations for at least one edge.

The Rashomon property: Markov equivalence. The key structural fact is that Markov equivalence classes are non-trivial for nearly all graph structures of practical interest.

Definition 17 (Markov Equivalence)**.** *Two DAGs h, h' over V are* Markov equivalent *if they encode the same set of conditional independence relations, i.e.,* $\mathsf{obs}(h) = \mathsf{obs}(h')$*. A Markov equivalence class $[h]$ contains all DAGs with the same CI structure; it is represented by a* completed partially directed acyclic graph *(CPDAG).*

Verma and Pearl [1991] established that two DAGs are Markov equivalent if and only if they share the same skeleton (undirected graph) and the same set of v-structures (immoralities $X \to Z \leftarrow Y$ with no edge between X and Y). Consequently, any undirected edge in the CPDAG corresponds to a reversible orientation: there exist two Markov-equivalent DAGs that orient it in opposite directions. Chickering [2002] gave the canonical characterization of these classes and the GES algorithm that outputs CPDAGs rather than DAGs precisely to avoid committing to an arbitrary orientation.

Proposition 18 (Causal Rashomon Property)**.** *$\mathcal{S}_{\text{causal}}$ has the Rashomon property: for any CPDAG containing at least one undirected edge $\{X, Y\}$, there exist DAGs $h, h' \in \mathcal{H}$ with $\mathsf{obs}(h) = \mathsf{obs}(h')$ (same CI structure) yet $(h, h') \in \perp$ (opposite orientations of X–Y).*

Proof. Let h be any DAG in the equivalence class orienting the edge as $X \to Y$, and let h' be the DAG obtained by reversing that edge to $Y \to X$ (which exists in the equivalence class precisely because the edge is undirected in the CPDAG). By Markov equivalence, $\mathsf{obs}(h) = \mathsf{obs}(h')$. By definition of $\perp$, $(h, h') \in \perp$. □

This is the formalization captured in `CausalInstance.lean` and `CausalDiscovery.lean`: the `rashomon` field of `causalSystem` is exactly the witness pair (h, h') from Proposition 18.

Instance 5: causal discovery impossibility.

Theorem 19 (Causal Discovery Impossibility — Instance 5)**.** *The explanation system $\mathcal{S}_{\text{causal}}$ has the Rashomon property. Therefore, by Theorem 6, no orientation rule exp can simultaneously be:*

1. ***Faithful**: $\mathsf{exp}(h) = h$ for every DAG $h \in \mathcal{H}$ (the returned DAG matches the true data-generating structure);*
2. ***Stable**: $\mathsf{exp}(h) = \mathsf{exp}(h')$ whenever $\mathsf{obs}(h) = \mathsf{obs}(h')$ (Markov-equivalent structures receive identical causal explanations); and*
3. ***Decisive**: exp always commits to a single fully oriented DAG (never returns a partial orientation or equivalence class).*

Proof. By Proposition 18, $\mathcal{S}_{\text{causal}}$ has the Rashomon property. Apply Theorem 6. □

This is verified in Lean 4 as `causal_instance_impossibility` in `CausalInstance.lean` (three axioms: `CausalConfig`, `CausalExplanation`, `CausalObservable`), and as the self-contained `causal_discovery_impossibility` in `CausalDiscovery.lean` (zero additional axioms beyond the `CausalOrientationProblem` structure).

Connection to the classical Markov equivalence problem. Theorem 19 is the classical Markov equivalence underdetermination recast in the universal framework. The three properties above correspond exactly to the three goals that any "complete" causal discovery method would need to achieve: return the true DAG (faithfulness), be invariant to observationally equivalent parameterizations (stability), and produce a unique answer (decisiveness). The Markov equivalence theorem of Verma and Pearl [1991] has long been understood to block this combination; our contribution is to show it is a *special case* of the single four-line proof of Theorem 6.

Resolution: CPDAGs as G-invariant explanations. The uniform resolution of Section 4 specializes to causal discovery as follows. Let $G = [h]$ denote the Markov equivalence class (the symmetry group acting by edge reversals on the set of reversible edges). The G-invariant explanation map aggregates over all member DAGs:

$$\mathsf{exp}^*(h) \;=\; \mathrm{CPDAG}([h]),$$

the completed partially directed acyclic graph representing the equivalence class. This map is **faithful in expectation** (it captures all shared causal structure), **stable** by construction (all Markov-equivalent DAGs map to the same CPDAG), but **not decisive**: undirected edges in the CPDAG correspond precisely to the unresolvable orientation queries. Reporting the CPDAG is therefore the principled response to the impossibility—trading decisiveness for faithfulness and stability, exactly as predicted by Theorem 6. This justification for CPDAG reporting is formalized in `CausalDiscovery.lean` as `cpdag_is_stable`.

3.6 Instance 6: Model Selection Under the Rashomon Effect

Model selection—choosing a single best model from a trained set—is perhaps the most commonplace decision in applied machine learning. Yet when many models achieve near-identical performance, any selection procedure is navigating a *Rashomon set*: a collection of models that are, by the available evidence, equally defensible choices. We show that this setting is an instance of Theorem 6, and that the resulting instability is not a failure of any particular selection criterion but a mathematical inevitability.

Explanation system. We instantiate the abstract framework as follows.

Definition 20 (Model Selection Explanation System)**.** *The* model selection explanation system *is the tuple* $\mathcal{S}_{\mathrm{sel}} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ *where:*

- $\mathbf{\Theta} = \mathbf{\Theta}$ *is the space of* model configurations *(hyperparameter settings, architecture choices, training seeds, or any parameterization of the model class under comparison);*
- $\mathcal{H}$ *is the* hypothesis space *of trained models; each* $h \in \mathcal{H}$ *corresponds to a configuration* $\theta(h) \in \mathbf{\Theta}$ *via the attribution map* $\theta : \mathcal{H} \to \mathbf{\Theta}$*;*
- Y *is the space of* performance metrics *(validation loss, accuracy, AUC, or any scalar summary of model quality on a held-out evaluation set); formally,* Y *is equipped with a strict total order* $>$ *representing "performs better than";*
- $\mathsf{obs} : \mathcal{H} \to Y$ *maps each hypothesis to its* observed performance metric *on the evaluation set;*
- $\mathsf{exp} : \mathcal{H} \to Y$ *is the* selection rule*—the criterion used to rank or choose among models; and*
- $\perp \subseteq Y \times Y$ *is defined by* $(y, y') \in \perp$ *whenever* $y \neq y'$ *and the two metrics imply opposite selection decisions (i.e., the model recommended under* y *is strictly disfavored under* y'*).*

The Rashomon property for model selection. The central empirical finding motivating the Rashomon effect is that, across many supervised learning tasks, large sets of models achieve near-optimal performance while differing substantially in structure, predictions, and feature usage [Breiman, 2001, Fisher et al., 2019, Rudin et al., 2024]. Formally, define the *ε-Rashomon set* as

$$\mathcal{R}_\varepsilon = \left\{ h \in \mathcal{H} : \mathsf{obs}(h) \geq \mathsf{obs}(h^*) - \varepsilon \right\},$$

where h^* is the empirical risk minimizer and $\varepsilon > 0$ is a tolerance. The Rashomon property holds for $\mathcal{S}_{\text{sel}}$ whenever $\mathcal{R}_\varepsilon$ contains pairs of models that are (i) indistinguishable in performance ($\theta(h) = \theta(h')$ in the sense that they share the same configuration class) yet (ii) produce incompatible selection metrics on different evaluation samples.

Proposition 21 (Rashomon Property for Model Selection)**.** *If the ε-Rashomon set $\mathcal{R}_\varepsilon$ contains two models $h, h' \in \mathcal{H}$ with $\theta(h) = \theta(h')$ such that $\mathsf{obs}(h) \neq \mathsf{obs}(h')$, then $\mathcal{S}_{\text{sel}}$ satisfies the Rashomon property (Definition 5).*

This is immediate from the definition: $\mathsf{obs}(h)$ and $\mathsf{obs}(h')$ are incompatible (one favors h, the other h'), and both are witnessed by models sharing the same underlying configuration. The condition is generically satisfied: Fisher et al. [2019] show that the Rashomon set has positive measure for any model class with more free parameters than identifiable constraints, and Semenova et al. [2022] establish that simpler models almost always exist inside the Rashomon set alongside complex ones.

Lean formalization. The model selection impossibility is mechanically verified in two Lean 4 files. `ModelSelection.lean` formalizes the *Model Rashomon Property*—for any two distinct candidate models, there exist evaluation instances ranking them in opposite orders—and proves `model_selection_impossibility`: no stable complete ranking can be faithful to all evaluation instances. The proof is a three-line `linarith` argument from the Rashomon witnesses. `ModelSelectionInstance.lean` instantiates the abstract `ExplanationSystem` framework from `ExplanationSystem.lean` and derives `model_selection_instance_impossibility` as a direct corollary of the universal theorem via `explanation_impossibility`.

Instance 6 (Model Selection Impossibility).

Theorem 22 (Model Selection Impossibility)**.** *The model selection explanation system $\mathcal{S}_{\text{sel}}$ (Definition 20) satisfies the Rashomon property whenever the ε-Rashomon set has positive measure. Consequently, by Theorem 6, no selection rule $\mathsf{exp} : \mathcal{H} \to Y$ can simultaneously be:*

1. ***Faithful:*** $\mathsf{exp}(h) = \mathsf{obs}(h)$ *for all h (the selected model is the one the evaluation metric actually endorses);*
2. ***Stable:*** $\mathsf{exp}(h) = \mathsf{exp}(h')$ *whenever $\theta(h) = \theta(h')$ (the same configuration always yields the same selection); and*
3. ***Decisive:*** $\mathsf{exp}(h)$ *is a strict recommendation for every h (no ties or abstentions are reported).*

Design space. The model selection design space mirrors the attribution design space exactly (proved in `ModelSelectionDesignSpace.lean`): every selection rule falls into one of two families.

- **Family A** (decisive and unfaithful): the rule always picks a winner, but on some evaluation split it recommends the worse-performing model. Standard single-criterion selection rules (e.g., argmax validation accuracy) fall here whenever the Rashomon set is non-trivial.
- **Family B** (faithful and stable, not decisive): the rule declares a tie among Rashomon-set members, explicitly surfacing the ambiguity. This corresponds to reporting the Rashomon set $\mathcal{R}_\varepsilon$ rather than a single selected model.

Implications. Theorem 22 formalizes and sharpens the empirical observation of Fisher et al. [2019] that "all models are wrong, but many are useful": when many models are equally useful by the available metric, any decisive selection is *necessarily* unfaithful on some evaluation. The instability practitioners observe when re-running AutoML pipelines, changing random seeds, or switching validation folds is not a software bug—it is the Rashomon property in action.

The resolution is the same as for feature attribution (Section 3.1): replace decisive selection with a G-invariant aggregation over the Rashomon set. Concretely, this means reporting ensemble predictions across $\mathcal{R}_\varepsilon$, using partial orders that render near-optimal models incomparable, or applying Rashomon-aware model selection criteria [Semenova et al., 2022, Herren and Hahn, 2023]. Each of these trades decisiveness for the pair (faithful, stable), which Theorem 6 shows is the only achievable combination under underspecification.

4 The Universal Resolution

5 Ubiquity

6 Lean Formalization

The Universal Explanation Impossibility and all six instances are mechanically verified in Lean 4 [de Moura and Ullrich, 2021] using Mathlib [The mathlib Community, 2020]. Table 1 summarizes the formalization.

Table 1: Summary of the Lean 4 formalization.

Component	Count	Notes
Files	67	All in `UniversalImpossibility/`
Theorems & lemmas	319	Verified by `grep -c`
Axioms	60	Domain-specific; core theorem uses 0
`sorry`s	0	Zero admitted goals
Core impossibility	1 theorem	Zero axiom dependencies
GBDT instance	4 axioms	Surjectivity, first-mover, split structure
DASH resolution	6 axioms	+ cross-group symmetry
Causal instance	3 axioms	CPDAG Markov equivalence
Neural net instance	2 axioms	Overparameterization + symmetry
Fairness instance	2 axioms	Unequal base rates + equal accuracy

The core impossibility (`universal_explanation_impossibility` in `UniversalImpossibility/Universal.lean`) depends on zero axioms beyond the Rashomon property, which is stated as a hypothesis rather than an axiom. Each instance introduces the minimum axioms needed to verify that its particular explanation system satisfies the Rashomon property.

The Lean proof of Theorem 6 is four lines, mirroring the paper proof exactly. The formalization also includes:

- The G-invariant resolution (Section 4): `Corollary.lean`, `DesignSpace.lean`;
- Quantitative bounds for GBDT (`Ratio.lean`), Lasso (`Lasso.lean`), and neural networks (`NeuralNet.lean`);
- The fairness instance (`FairnessAudit.lean`) and causal instance (`CausalDiscovery.lean`).

7 Discussion: What Explainability Can and Cannot Do

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